Covariance, Correlation, and OLS Assumption 1

Wenjie Wang

Economics, School of Social Sciences
Nanyang Technological University

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The (population) covariance between X and Y is

$$\sigma_{XY} = Cov(X, Y) = E\left[(X - \mu_X)(Y - \mu_Y)\right]$$

where $\mu_X = E(X)$ and $\mu_Y = E(Y)$. From the definition we can see $\sigma_{XY} > 0$ means that **on average**, when X is above its mean (μ_X) Y is also above its mean (μ_Y): X and Y tend to move in the **same direction**. On the other hand, $\sigma_{XY} < 0$ means that **on average**, when X is above its mean Y is below its mean: X and Y tend to move in the **opposite direction**. Also note that the covariance is a measure of **linear association** between X and Y, but it is not a measure for nonlinear association.

In practice, σ_{XY} is estimated by the **sample covariance**:

$$s_{XY} = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y}),$$

where dividing by $n - 1$ instead of n is called a **degree-of-freedom correction**: estimating μ_X and μ_Y by $\bar{X}$ and $\bar{Y}$ already used some information in the data.

The (population) correlation between X and Y is a **standardized version** of the covariance:

$$corr(X,Y) = \frac{Cov(X,Y)}{\sqrt{Var(X)Var(Y)}} = \frac{\sigma_{XY}}{\sqrt{\sigma_X^2 \sigma_Y^2}}.$$

The value of the **correlation is between -1 and 1**: $-1 \leq corr(X,Y) \leq 1$. Therefore, it is easier to use in practice than the covariance.

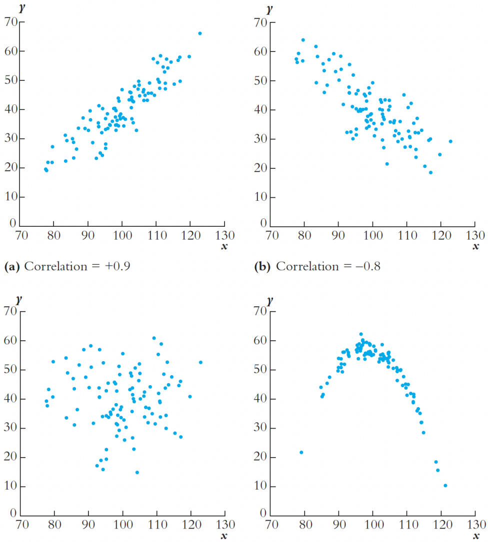
In practice, $corr(X,Y)$ is estimated by the **sample correlation**:

$$r_{XY} = \frac{s_{XY}}{\sqrt{s_X^2 s_Y^2}},$$

where $s_X^2 = \frac{1}{n-1}\sum_{i=1}^n (X_i - \bar{X})^2$ is the sample variance of X, and $s_Y^2 = \frac{1}{n-1}\sum_{i=1}^n (Y_i - \bar{Y})^2$ is the sample variance of Y.

FIGURE 3.3 Scatterplots for Four Hypothetical Data Sets

The scatterplots in Figures 3.3a and 3.3b show strong linear relationships between X and Y. In Figure 3.3c, X is independent of Y and the two variables are uncorrelated. In Figure 3.3d, the two variables also are uncorrelated even though they are related nonlinearly.



(a) Correlation = +0.9

(b) Correlation = −0.8

(c) Correlation = 0.0

(d) Correlation = 0.0 (quadratic)

The correlation can be used to check whether OLS Assumption 1 of the linear regression model $Y = \beta_0 + \beta_1 X + u$ holds for our data set.

In particular, if $corr(X, u) \neq 0$ (which also implies $Cov(X, u) \neq 0$), then $E(u|X) \neq 0$. That is, **if X and u are positively or negatively correlated, then OLS Assumption 1 is violated.**

The formal proof is provided in the Appendix, but there is no need to recite it.

For example, suppose we consider $wage = \beta_0 + \beta_1 educ + u$, where the error term includes *abil*. OLS Assumption 1 is very unlikely to hold because we know that typically **people with higher** *abil* **also have higher** *educ* (*abil* **and** *educ* **are positively correlated**). Thus, it is not a good idea to run a simple linear regression of *wage* on *educ*.

For the example of test scores and class sizes in primary schools:

$$TestScore = \beta_0 + \beta_1 STR + u. \tag{1}$$

(1) The kids are randomly enrolled to primary schools.
(2) The kids are selectively enrolled: the kids with high ability will be enrolled to good schools and those with low ability will be enrolled to bad schools.
(3) The kids' parents decide which schools to apply. In the case that the number of applicants exceeds the vacancy of the school, a random balotting will be implemented.

For (1), the random assignment of kids implies that *STR* and *abil* will be uncorrelated.

For (2), the selection with regard to kids' abilities implies that *abil* and *STR* will be negatively correlated (on average *abil* is low when *STR* is high, and vice versa). Therefore, **OLS Assumption 1 does not hold** in this case.

For (3), assuming all parents will apply for the good schools first (assuming they all care about their kids), kids' with different levels of ability will have the same chances to be enrolled into good schools and bad schools, *STR* and *abil* will be uncorrelated in this case.

Appendix.

$E[u|X] = 0$ implies $Cov(X, u) = 0$ and $corr(X, u) = 0$, where $Cov(X, u)$ denotes the covariance between X and u, and $corr(X, u)$ denotes the correlation between X and u.

Proof.

First note that

$$\begin{aligned} Cov(X,u) &= E[(X - E(X))(u - E(u))] \\ &= E[Xu - XE(u) - E(X)u + E(X)E(u)] \\ &= E(Xu) - E(X)E(u) = E(Xu), \end{aligned} \tag{2}$$

where the last equality follows from the fact we can always assume $E(u) = 0$.

[**NO Need** to recite the following.]
Second, we note that

$$E(Xu) = E[E(Xu|X)] = E[XE(u|X)] = E[X \cdot 0] = 0, \qquad (3)$$

where the first equality follows from the **Law of Iterated Expectation**, the second equality follows from the **property of conditional expectation**, and the third equality follows from $E[u|X] = 0$.

Combining eqs.(2) and (3), we conclude that $E[u|X] = 0$ implies $Cov(X, u) = 0$. In addition, $Cov(X, u) = 0$ implies $corr(X, u) = 0$, and this completes the proof. ■